Together First, Apart Later? When Optimizers Start to Learn Different Functions

Initial checkpoint, MIT 6.7910, Fall 2026

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Summary. We train one transformer from one initialization with AdamW and with Muon (Jordan et al., 2024), on the same mini-batches. (1) At equal training loss, AdamW and Muon predict as similarly as two runs of one optimizer on different data orders; the loss barrier between their final weights exceeds data-order barriers in only one of three settings. (2) The singular values of their weight and update matrices differ beyond data-order variation (Muon's more evenly sized, AdamW's dominated by a few large ones), while their prediction statistics do not. (3) A run switched to the other optimizer mid-training ends, in its predictions, no further from either optimizer's own run than two data orders are; its updates take the new optimizer's form, but its weights and activations keep part of the old one's.

Setup. The main setting, TinyStories-10M, trains a 14M-parameter nanoGPT (Karpathy, 2022) (10.6M without embeddings and head) on TinyStories (Eldan and Li, 2023) for 10,000 steps of 128 sequences × 256 tokens with a cosine learning-rate decay; two more settings, a 30M model on TinyStories (TinyStories-30M) and a character-level model on Shakespeare, serve as checks (Appendix A). Muon orthogonalizes the update of each weight matrix in the blocks (sets its singular values to about 1); a separate AdamW trains the embeddings, output head and LayerNorm gains; each optimizer uses the best of five learning rates tried. All runs start from the same initial weights. A *data order* is a fixed choice of mini-batch for every step; each optimizer is run on five. The distance between two runs is the mean Jensen–Shannon divergence (JS, in bits) of their next-token predictions on 2,048 fixed held-out positions. We compare two runs at the checkpoints where they reach the same *training loss* (on 64 fixed training sequences), so that a faster optimizer is not counted as different, and divide their distance by a *floor*: the mean distance between two data orders of one optimizer (for two optimizers, the larger of their floors). A distance of about one floor thus means "as different as two data orders". For any other metric, we compare two optimizers by the *ratio* of their values at equal training loss, with a margin over data orders: if the first optimizer's values range over $[a_{\min}, a_{\max}]$ and the second's over $[b_{\min}, b_{\max}]$, the ratio lies in $[a_{\min}/b_{\max},\ a_{\max}/b_{\min}]$, written as midpoint ± half-width; an interval that excludes 1 means the optimizers differ by more than data orders do. The *loss barrier* between two weight vectors is the largest amount, over 11 evenly spaced points on the straight line between them, by which the training loss exceeds the straight line joining its two end values (Frankle et al., 2020); with a near-zero barrier they are *linearly connected*. Losses and barriers are in nats. In a *swap*, a run of one optimizer (the *parent*, data order 0) is switched to the other at a *fork step* and trained to step 10,000 on the parent's batches; the new optimizer starts from a *warm state*, its momentum statistics gathered alongside the parent at learning rate 0. Its *own run* is the run trained with it from the start. In total we trained 141 runs using 33 GPU-hours (Appendix D).

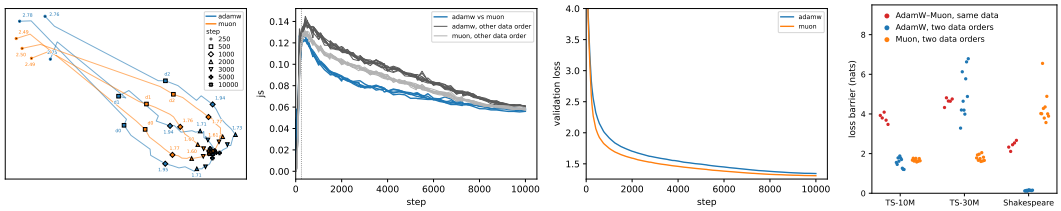


Figure 1: AdamW (blue) and Muon (orange); first three panels on TinyStories-10M. First: checkpoints from step 250 on (data orders d0–d2), placed in 2-D by densMAP from $\sqrt{\text{JS}}$; numbers: training loss at steps 250, 1000, 2000. The map shows which checkpoints are nearest, not how far apart they are: the six endpoints, drawn nearly overlapping, are JS 0.056–0.063 apart, about as far as each run moves between steps 5000 and 10,000. Second: JS between AdamW and Muon on the same data order at the same step (blue) and between two data orders of one optimizer (grey); dotted: step 250. Third: validation loss (mean, range over data orders). Right: loss barrier between two final weight vectors (how far the training loss rises along the straight path between them above the line joining the two end losses), one point per pair: AdamW and Muon on the same data order (red, 5 pairs), two data orders of AdamW (blue) or Muon (orange; 10 pairs each).

Step 1: how the trajectories differ (Fig. 1). After the learning-rate warmup, AdamW and Muon runs on the same data order stay close: each such pair are nearest neighbours in the densMAP plot (Fig. 1, left), and at every step their JS is smaller than between two data orders of one optimizer (second panel). Muon learns faster (third panel): by step 7500 it reaches AdamW's final validation loss, 1.344. Compared at equal training loss, the AdamW–Muon distance is about equal to the floor: at training losses below 4.85 it is 0.95–1.38 times the floor (median 1.02), and 0.99 times at the lowest loss all runs reach (1.02 on each of TinyStories-30M and Shakespeare). So, judged by their predictions, AdamW and Muon end as similar as two runs of one optimizer on different data orders.

Their final weights, however, are further apart than two data orders' are, at least on TinyStories-10M (Fig. 1, right): the loss barrier is 3.5–4.1 between AdamW and Muon, against 1.2–1.9 between two data orders of AdamW and 1.6–1.8 between two of Muon. On TinyStories-30M and Shakespeare, the mean AdamW–Muon barrier lies between the two optimizers' mean data-order barriers (4.64 between 1.78 and 5.05; 2.42 between 0.14 and 4.34).

Step 2: what differs along the way. We measure 30 properties of each model at 34 points, from the first steps to the final model (Appendix B). They form five groups: the weights, the change made by one update step, the loss curvature, internal activations, and the predictions. We compare optimizers by each metric's ratio at training loss 1.81, with three data orders each (Fig. 2; AdamW and Muon in Table 4; SGD with momentum as a reference in Table 5).

All four metrics of the weight matrices' singular values (spectral norm, stable and effective rank, singular-value spread) differ between AdamW and Muon beyond data-order variation (Table 4), while the prediction metrics (entropy, confidence, calibration, accuracy, losses on frequent and rare targets, JS to n-gram models) all have AdamW/Muon intervals containing 1, all but calibration error's within 1 ± 0.03, as does how far the weights move (1.01 ± 0.02). Muon's weight matrices have more even singular values: their stable rank $\|W\|_F^2/\|W\|_2^2$ is 2.61 ± 0.04 times AdamW's (86 against 33), and AdamW's largest-to-median singular-value ratio $\sigma_1/\sigma_{\text{median}}$ is 2.98 ± 0.03 times Muon's (8.0 against 2.7). AdamW's model uses fewer MLP units, has larger activations and sharper attention, and sits where the loss is less curved; relative to Muon's, its fraction of active MLP units is 0.539 ± 0.007 times as large (12% against 22%), its residual-stream RMS 1.40 ± 0.03 times, its attention entropy 0.899 ± 0.014 times, and its Hessian trace $\operatorname{tr} H$ 0.510 ± 0.019 times (top eigenvalue 0.433 ± 0.013 times). Muon's one-step updates have a stable rank 13.1 ± 0.2 times AdamW's (220 against 17), by construction. SGD differs from both (Table 5): AdamW and Muon end 4.47 ± 0.11 and 4.42 ± 0.07 times as far from the initialization, take relative steps 10.9 ± 1.0 and 10.8 ± 0.9 times as large, less aligned with the batch gradient (cosine 0.24, 0.22 against 0.45), and SGD's $\operatorname{tr} H$ is 18.0 ± 0.8 and 9.18 ± 0.20 times theirs.

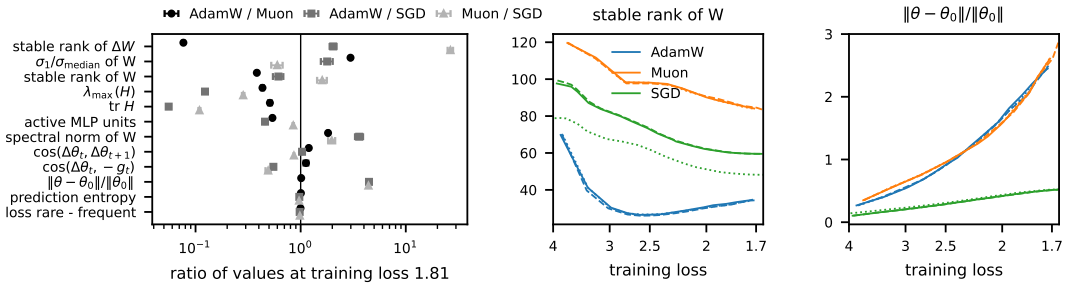


Figure 2: TinyStories-10M, three data orders per optimizer. Left: ratio of two optimizers' values at training loss 1.81 for selected metrics (log scale; bars: margins; all metrics in Tables 4 and 5). Δ: change over one step; g_t: gradient on the step's batch; H: Hessian on 32 training sequences. Middle, right: stable rank of W and $\|\theta - \theta_0\|/\|\theta_0\|$ against training loss; colour: optimizer, line style: data order.

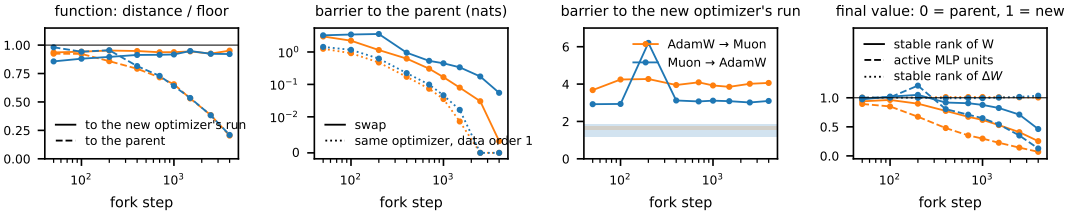


Figure 3: Swaps on TinyStories-10M, data order 0, against the fork step; each panel shows a property of the swapped run's final model. Left: JS from the swapped run to the new optimizer's own run and to the parent, each divided by the floor of that run's optimizer. Second: loss barrier to the parent, for the swapped run and, as a control, for the parent's own optimizer continued from the same step on data order 1. Third: loss barrier to the new optimizer's own run; bands: barriers between two data orders of that optimizer. Right: stable rank of W and of ΔW and fraction of active MLP units, rescaled so that 0 is the parent's final value and 1 the final value of the new optimizer's own run.

Step 3: swaps (Fig. 3). We swap at nine fork steps from 50 to 4000, in both directions (Appendix C). Every swapped run ends closer than two data orders are to both the new optimizer's own run (0.86–0.95 of the floor) and the parent (0.92–0.98 at fork step 50, 0.20–0.21 at 4000): in its predictions it is as similar to either run as two data orders are. Its weights are not. It is linearly connected to its parent (barrier below 0.05) only from fork step 2500 when leaving AdamW and never when leaving Muon, against 1000 for a switch to data order 1, and never as close to the new optimizer's own run as that optimizer's data orders are to each other (third panel). In the final model (right), the one-step updates take the new optimizer's form at every fork step (stable rank of ΔW 0.99–1.04 of the way from the parent's value to the new optimizer's), while the weights and activations keep part of the parent's, the more the later the swap: at fork step 4000, the stable rank of W moves 0.25 (AdamW to Muon) and 0.46 (Muon to AdamW) of the way, and the fraction of active MLP units 0.07 and 0.13.

Next. (i) Steps 2 and 3 on TinyStories-30M and Shakespeare; (ii) do the weight spectra explain the AdamW–Muon barriers? (iii) talk (Dec 1–4); paper, code, Colab (Dec 5–11). https://github.com/qqgjyx/MIT-6.7910

References

Eldan, R. and Y. Li (2023). *TinyStories: How Small Can Language Models Be and Still Speak Coherent English?*

Frankle, J., G. K. Dziugaite, D. M. Roy, and M. Carbin (2020). "Linear Mode Connectivity and the Lottery Ticket Hypothesis". In: *ICML*.

Jordan, K., Y. Jin, V. Boza, et al. (2024). *Muon: An Optimizer for Hidden Layers in Neural Networks*. Blog post. URL: https://kellerjordan.github.io/posts/muon/.

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A Hyperparameters

Table 1: Settings shared by the AdamW and Muon runs on TinyStories-10M. Values from the repository's configuration files and code.

Runs	AdamW and Muon × data orders d0–d4 (10)
Model	nanoGPT, pre-LayerNorm blocks
Layers, heads, width	6, 6, 384
MLP	GELU, width 1536
Biases, dropout	none, none
Output head	untied
Attention projections	separate query, key, value
Context	256 tokens
Parameters	13.9M (10.6M without token and position embeddings and head)
Data	TinyStories (GPT-4 version)
Tokenizer	byte-level BPE, 4096 tokens, trained on the training split
Batch	128 windows × 256 tokens
Window offsets	from the training split; generator seeded with (data order, step), same for every optimizer
Steps	10,000
Tokens	328M (about 0.6 epochs)
Learning rate	$\eta\,\varphi(t)$ at step t, every parameter group
Warmup	linear, $\varphi(t) = (t+1)/200$ for $t < 200$
Decay	cosine to $\varphi = 0.1$ at step 10,000
Gradient clipping	none
Initialization seed	0, all runs
Training precision	bfloat16 autocast
Prediction, Hessian precision	float32, no autocast (TF32 matrix products allowed)
Logged predictions	next-token distributions: 128 fixed validation sequences × 16 evenly spaced positions (2,048), as float16 log-probabilities
Training loss	64 fixed training sequences, all positions; used to match runs at equal loss
Both logged at	every 250th step + 30 log-spaced steps in $[1, 10{,}000]$
Hessian	top eigenvalue, 20 Lanczos iterations, 32 fixed training sequences; AdamW also of $P^{-1/2}HP^{-1/2}$, with P AdamW's diagonal preconditioner
Measured at	every 500th step + 30 log-spaced steps
Loss barrier	64 fixed training sequences, 11 evenly spaced points of the segment between the final weights

Table 2: Optimizer-specific settings of the TinyStories-10M runs. n_{out}, n_{in}: output and input size of the matrix.

	AdamW	Muon
η	$2 \cdot 10^{-3}$	0.01
η choice	best final validation loss of a 5-rate sweep on d0	as AdamW
Update	bias-corrected moments	Nesterov momentum; 5 quintic Newton–Schulz steps in bfloat16
Momentum	$\beta = (0.9, 0.95)$	$\beta = 0.95$
ϵ	10^{-8}	–
Update scale	–	$\sqrt{\max(1, n_{\text{out}}/n_{\text{in}})}$
Weight decay	0.1, decoupled; none on LayerNorm gains	none
Parameters	all	query, key, value, attention-output and both MLP matrices of every block
Aux. AdamW	–	embeddings, head, LayerNorm gains
Aux. η, β, ϵ	–	$2 \cdot 10^{-3}$, $(0.9, 0.95)$, 10^{-8}
Aux. weight decay	–	none

Table 3: Shakespeare-char and TinyStories-30M: rows that differ from TinyStories-10M; all else as in Tables 1 and 2. "Aux.": Muon's auxiliary AdamW.

	Shakespeare-char	TinyStories-30M
Runs	AdamW, Muon × d0–d4 (10)	AdamW, Muon × d0–d4 (10)
Data	Shakespeare, nanoGPT's 90/10 split	TinyStories
Vocabulary	65 characters	4096 (BPE)
Layers, heads, width	6, 6, 384	8, 8, 512
Parameters (without embeddings, head)	10.8M (10.6M)	29.5M (25.2M)
Batch	64×256	128×256
Steps	1,500	20,000
Tokens	24.6M (about 24 passes over the 1M training characters)	655M (about 1.2 epochs)
Warmup steps	100	400
Cosine decay to step	1,500	20,000
η_{AdamW}	$3 \cdot 10^{-4}$	$2 \cdot 10^{-3}$
η_{Muon} (aux.)	0.04 (10^{-3})	0.01 ($1.5 \cdot 10^{-3}$)
η chosen by	minimum validation loss over training (runs overfit)	best final validation loss
AdamW β	$(0.9, 0.99)$, as nanoGPT	$(0.9, 0.95)$
Predictions	all 256 positions of 64 validation sequences (16,384)	16 positions of 128 sequences (2,048)
Predictions at	every 50th + 40 log-spaced	every 500th + 30 log-spaced

B Metrics

The Step-2 runs save checkpoints every 1000 steps, at 12 log-spaced steps, and at three consecutive steps $t, t+1, t+2$ for 24 values of t from 25 to 9998 (to measure one-step updates); we measure the metrics at 34 of these per run, and the Step-3 swaps only at their final weights. *Weights*: for all parameters together, $\|\theta\|_2$, $\|\theta - \theta_0\|/\|\theta_0\|$ and $\|\theta\|_\infty/\|\theta\|_2$; and, averaged over the 36 weight matrices W of the blocks, the stable rank $\|W\|_F^2/\|W\|_2^2$, the effective rank $\exp H(\sigma/\sum\sigma)$ with H the entropy of the normalized singular values σ, the spectral norm, and $\sigma_1/\sigma_{\text{median}}$. *Updates* $\Delta\theta_t = \theta_{t+1} - \theta_t$: $\|\Delta\theta_t\|/\|\theta_t\|$, its cosine with $\Delta\theta_{t+1}$ and with the negative gradient $-g_t$ on step t's batch, and the stable and effective rank of its block matrices ΔW. *Curvature* of the training loss on 32 fixed training sequences, with Hessian H: $\lambda_{\max}(H)$ (20 Lanczos iterations, logged during training), $\operatorname{tr} H$ (10 Rademacher probes), $\operatorname{tr} H/\lambda_{\max}$, and the batch sharpness $\mathbb{E}_B[g_B^\top H_B g_B]/\mathbb{E}_B\|g_B\|^2$, where g_B and H_B are the gradient and Hessian on a mini-batch B. *Predictions*, on the 2,048 held-out positions: entropy, mean top-1 probability, calibration error (15 bins), top-1 accuracy, loss on the targets in the top and the bottom third of training-set frequency and their difference, and JS to the unigram and to the bigram model of the training split. *Activations*, on 32 held-out sequences: attention entropy and attention on the first token (means over layers, heads and positions), fraction of MLP pre-activations above 0, RMS of the residual stream after the last block, and effective rank of the centred final features.

Table 4: AdamW and Muon at training loss 1.81 on TinyStories-10M (settings in Tables 1 and 2): all metrics of Step 2 (Fig. 2). Columns AdamW and Muon: mean ± half the range over data orders 0–2, in units of 10^k in rows marked ($\times 10^k$). Column A/M: AdamW's value divided by Muon's; with $a_{\min}, a_{\max}$ and $b_{\min}, b_{\max}$ their extremes over data orders, the ratio lies in $[a_{\min}/b_{\max},\ a_{\max}/b_{\min}]$, given as midpoint ± half-width, and an interval that excludes 1 means a difference beyond data-order variation. Bold: the five ratios farthest from 1 (on a log scale) among those whose interval excludes 1.

	AdamW	Muon	A/M
Weights			
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	0.997 ± 0.013
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	1.01 ± 0.02
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	1.09 ± 0.01
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	$\mathbf{0.383 \pm 0.005}$
effective rank of W	272 ± 0	325 ± 0	0.839 ± 0.000
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.82 ± 0.03
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	$\mathbf{2.98 \pm 0.03}$
Updates (one step)			
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	1.01 ± 0.02
stable rank of ΔW	16.7 ± 0.3	220 ± 0	$\mathbf{(7.66 \pm 0.14) \times 10^{-2}}$
effective rank of ΔW	228 ± 1	369 ± 0	0.618 ± 0.003
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	1.19 ± 0.01
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	1.12 ± 0.06
Curvature			
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	$\mathbf{0.433 \pm 0.013}$
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	$\mathbf{0.332 \pm 0.116}$
Hessian trace	1400 ± 40	2770 ± 20	0.510 ± 0.019
trace / top eigenvalue	224 ± 6	196 ± 7	1.16 ± 0.08
Activations			
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	0.899 ± 0.014
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	0.908 ± 0.079
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.539 ± 0.007
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	1.40 ± 0.03
effective rank of features	312 ± 2	320 ± 0	0.977 ± 0.007
Predictions			
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.01 ± 0.01
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.994 ± 0.007
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	1.09 ± 0.22
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	1.01 ± 0.02
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	1.00 ± 0.01
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	1.00 ± 0.01
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	0.996 ± 0.015
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.998 ± 0.002
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.996 ± 0.004

Table 5: The metrics of Table 4 with SGD as a reference. SGD uses heavy-ball momentum 0.9 and learning rate 0.06, the only rate tried between 0.03 and 0.12 that did not diverge on any of the three data orders; everything else is as in Table 1. Columns AdamW, Muon and SGD: mean ± half the range over data orders 0–2, in units of 10^k in rows marked ($\times 10^k$). Columns A/S and M/S: AdamW's and Muon's value divided by SGD's, with the margin over data orders given as in Table 4. Bold: in each ratio column, the five ratios farthest from 1 (on a log scale) among those whose interval excludes 1.

	AdamW	Muon	SGD	A/S	M/S
Weights					
weight norm $\|\theta\|_2$	224 ± 2	224 ± 0	110 ± 0	2.04 ± 0.02	2.05 ± 0.01
distance from init. $/\|\theta_0\|$	2.18 ± 0.03	2.15 ± 0.01	0.486 ± 0.006	4.47 ± 0.11	4.42 ± 0.07
largest entry $/\|\theta\|_2$ ($\times 10^{-3}$)	7.04 ± 0.06	6.45 ± 0.02	39.5 ± 1.0	0.180 ± 0.006	$\mathbf{0.165 \pm 0.005}$
stable rank of W	33.0 ± 0.3	86.2 ± 0.3	56.2 ± 5.6	0.614 ± 0.069	1.60 ± 0.17
effective rank of W	272 ± 0	325 ± 0	321 ± 1	0.849 ± 0.003	1.01 ± 0.00
spectral norm of W	5.39 ± 0.07	2.95 ± 0.01	1.47 ± 0.11	3.61 ± 0.31	1.98 ± 0.15
$\sigma_1/\sigma_{\text{median}}$ of W	8.01 ± 0.05	2.68 ± 0.00	4.37 ± 0.55	1.79 ± 0.23	0.599 ± 0.074
Updates (one step)					
step size $\|\Delta\theta\|/\|\theta\|$ ($\times 10^{-3}$)	6.56 ± 0.08	6.53 ± 0.03	0.624 ± 0.047	$\mathbf{10.9 \pm 1.0}$	$\mathbf{10.8 \pm 0.9}$
stable rank of ΔW	16.7 ± 0.3	220 ± 0	8.37 ± 0.48	2.04 ± 0.15	$\mathbf{26.6 \pm 1.6}$
effective rank of ΔW	228 ± 1	369 ± 0	205 ± 2	1.12 ± 0.02	1.81 ± 0.02
cosine of consecutive steps	0.883 ± 0.002	0.738 ± 0.003	0.857 ± 0.004	1.03 ± 0.01	0.863 ± 0.007
cosine of step and $-g$	0.244 ± 0.004	0.217 ± 0.007	0.447 ± 0.010	0.550 ± 0.022	0.491 ± 0.027
Curvature					
top Hessian eigenvalue	6.18 ± 0.10	14.3 ± 0.2	50.2 ± 0.6	$\mathbf{0.123 \pm 0.004}$	0.284 ± 0.008
batch sharpness	0.974 ± 0.254	3.13 ± 0.37	6.38 ± 0.50	$\mathbf{0.162 \pm 0.051}$	0.504 ± 0.095
Hessian trace	1400 ± 40	2770 ± 20	$(2.55 \pm 0.04) \times 10^4$	$\mathbf{(5.55 \pm 0.25) \times 10^{-2}}$	$\mathbf{0.109 \pm 0.002}$
trace / top eigenvalue	224 ± 6	196 ± 7	507 ± 9	0.447 ± 0.020	0.385 ± 0.021
Activations					
attention entropy (nats)	2.51 ± 0.03	2.79 ± 0.02	3.84 ± 0.01	0.655 ± 0.009	0.728 ± 0.007
attention on first token ($\times 10^{-2}$)	1.75 ± 0.11	1.98 ± 0.05	1.82 ± 0.07	0.972 ± 0.095	1.07 ± 0.07
active MLP units	0.120 ± 0.001	0.222 ± 0.001	0.262 ± 0.002	0.458 ± 0.008	0.850 ± 0.012
residual-stream RMS	4.73 ± 0.05	3.38 ± 0.02	0.244 ± 0.003	$\mathbf{19.4 \pm 0.5}$	$\mathbf{13.8 \pm 0.3}$
effective rank of features	312 ± 2	320 ± 0	297 ± 2	1.05 ± 0.01	1.07 ± 0.01
Predictions					
prediction entropy (nats)	1.89 ± 0.01	1.88 ± 0.01	1.94 ± 0.00	0.971 ± 0.006	0.966 ± 0.007
mean top-1 probability	0.561 ± 0.003	0.563 ± 0.001	0.558 ± 0.001	1.00 ± 0.01	1.01 ± 0.00
calibration error ($\times 10^{-2}$)	3.25 ± 0.17	3.05 ± 0.46	2.97 ± 0.28	1.08 ± 0.16	1.02 ± 0.25
top-1 accuracy	0.546 ± 0.005	0.544 ± 0.005	0.554 ± 0.003	0.987 ± 0.013	0.980 ± 0.014
loss on frequent targets	0.896 ± 0.004	0.892 ± 0.008	0.851 ± 0.012	1.05 ± 0.02	1.05 ± 0.02
loss on rare targets	3.10 ± 0.02	3.10 ± 0.01	3.11 ± 0.04	0.999 ± 0.018	0.999 ± 0.015
rare minus frequent loss	2.21 ± 0.02	2.21 ± 0.01	2.26 ± 0.03	0.979 ± 0.020	0.982 ± 0.017
JS to unigram model	0.800 ± 0.001	0.802 ± 0.001	0.804 ± 0.001	0.995 ± 0.002	0.997 ± 0.003
JS to bigram model	0.469 ± 0.001	0.471 ± 0.001	0.462 ± 0.001	1.02 ± 0.00	1.02 ± 0.00

C Swaps

Table 6: Swap forks on TinyStories-10M. A: parent's optimizer; B: optimizer after the swap; s: fork step.

Parents	the d0 runs of AdamW and Muon (Table 1)
Shadow state	the other optimizer, stepped on the same mini-batch gradients with learning rate 0: momenta, second moments and step counts evolve as if training; parameters follow A alone
Fork steps s	50, 100, 200, 400, 700, 1000, 1500, 2500, 4000
Saved at s	parameters; states of both optimizers
Swap $A \to B$ loads	parameters at s; B's shadow state
Hyperparameters	B's (Table 2)
Steps	s to 10,000, on the parent's schedule and batches (data order 0)
Swap forks	2 directions × 9 steps = 18
Same-batch control	the parent itself (A continued with the same batches)
Data-order control	A continued from s with its own state, data order 1: 18 forks

D Compute

All runs used one NVIDIA GPU each on the MIT ORCD cluster. GPU-hours are those Slurm allocated to each suite (a group of runs launched together); where runs from several suites shared a GPU, its hours were split among them in proportion to training time. We did not count the measurements made after training (Muon's spectral-norm sharpness, loss barriers, distances and the metrics of Appendix B), which took a few GPU-hours.

Table 7: Runs described in the checkpoint and their compute.

	Runs	GPU-hours
Learning-rate sweeps, AdamW and Muon (3 settings)	30	7.1
Runs from the initialization, AdamW and Muon (3 settings, 5 data orders)	30	11.6
Swaps and data-order forks (Table 6)	36	7.1
Step 2: SGD learning-rate search, SGD runs at the chosen rate	$15 + 3$	3.3
Step 2: runs with consecutive checkpoints (3 optimizers $\times$ 3 data orders)	9	1.9
Step 3: swaps again, with checkpoints at steps 9998–10,000	18	2.1
Total	141	33.0